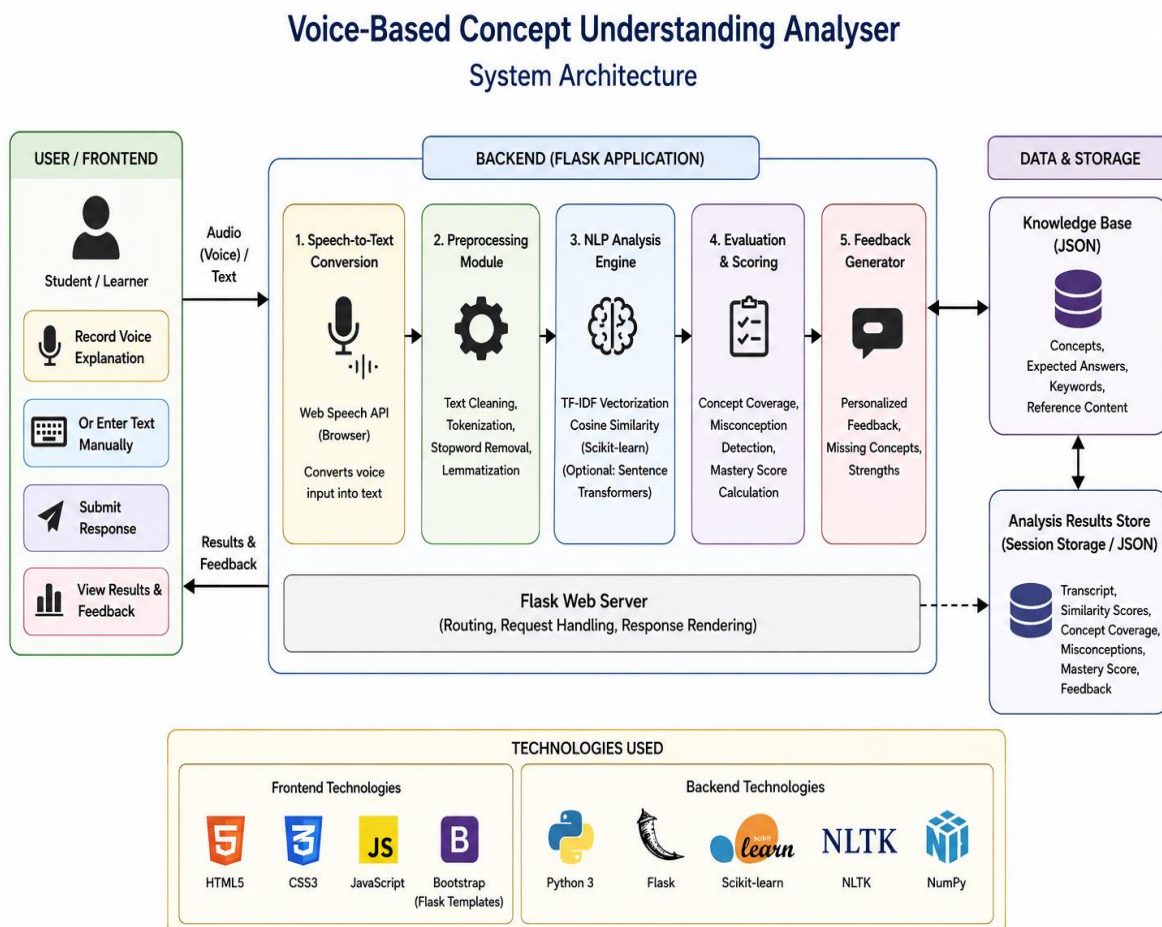


## Solution Architecture

|                      |                                            |
|----------------------|--------------------------------------------|
| <b>Date</b>          | 23 JUNE 2026                               |
| <b>Team ID</b>       | SWTID-2026-1377                            |
| <b>Project Name</b>  | Voice Based Concept Understanding Analyser |
| <b>Maximum Marks</b> | 5 Marks                                    |

### Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



## Component Description Table

| Component Name                         | Description / Role in Architecture                                                                                                                                                 | Technologies Used                                                                       |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| <b>User / Frontend</b>                 | Provides the interface for students to interact with the application. Users can record their voice, manually enter text, submit responses, and view analysis results and feedback. | HTML5, CSS3, JavaScript, Bootstrap, Flask (Jinja2 Templates)                            |
| <b>Speech-to-Text Module</b>           | Converts the student's spoken explanation into text using the browser's speech recognition functionality before NLP processing begins.                                             | Web Speech API                                                                          |
| <b>Text Preprocessing Module</b>       | Cleans and preprocesses the transcript by removing stop words, tokenizing text, and normalizing words to prepare the input for concept analysis.                                   | Python, NLTK                                                                            |
| <b>NLP Analysis Engine</b>             | Performs semantic similarity analysis between the student's explanation and reference concepts to evaluate conceptual understanding.                                               | Scikit-learn, TF-IDF Vectorization, Cosine Similarity, Sentence Transformers (optional) |
| <b>Evaluation &amp; Scoring Module</b> | Detects missing concepts and misconceptions, measures concept coverage, and calculates the student's mastery score.                                                                | Python, NumPy, Scikit-learn                                                             |
| <b>Feedback Generation Module</b>      | Generates personalized feedback, highlights strengths, identifies weak areas, and suggests improvements based on the analysis results.                                             | Python, Flask                                                                           |
| <b>Flask Web Server</b>                | Handles HTTP requests, routes pages, coordinates communication between frontend and backend modules, and returns evaluation results to the user.                                   | Flask (Python)                                                                          |
| <b>Knowledge Base</b>                  | Stores predefined concepts, expected answers, keywords, and reference explanations used for concept evaluation.                                                                    | JSON Files                                                                              |
| <b>Analysis Results Store</b>          | Temporarily stores transcripts, similarity scores, mastery scores, misconceptions, concept coverage, and generated feedback during user sessions.                                  | Session Storage, JSON                                                                   |

|                                   |                                                                                                                                 |                       |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <b>Output / Results Interface</b> | Displays the mastery score, personalized feedback, concept coverage, misconceptions, and overall analysis report to the student | HTML, CSS, JavaScript |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-----------------------|